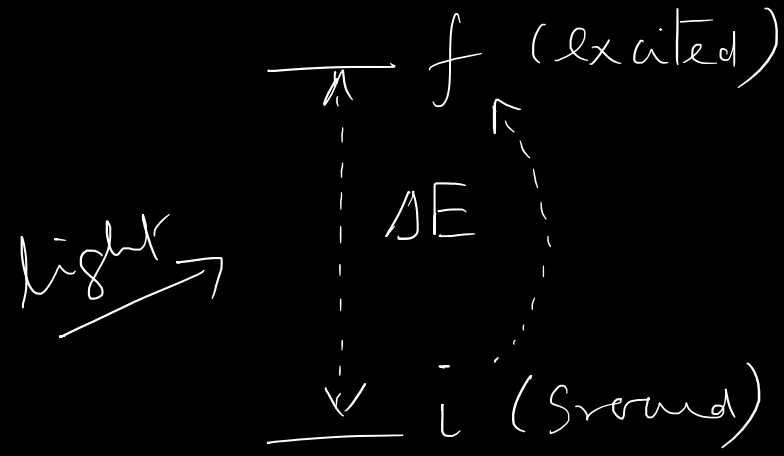
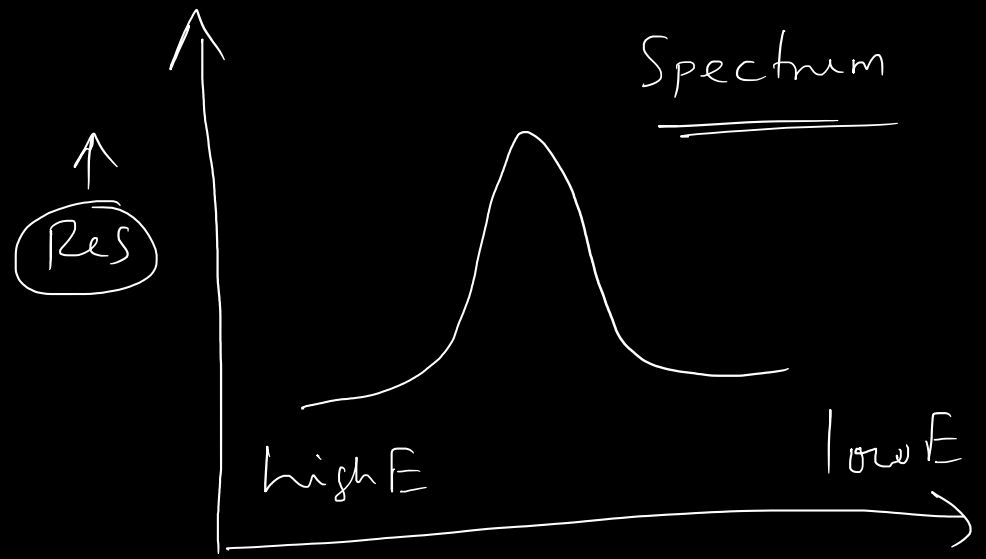
CH-1101: Molecular Spectroscopy



$$\Delta E = h\nu = \frac{hc}{\lambda} = hc\bar{\nu}$$

$$\Delta E = h\nu$$

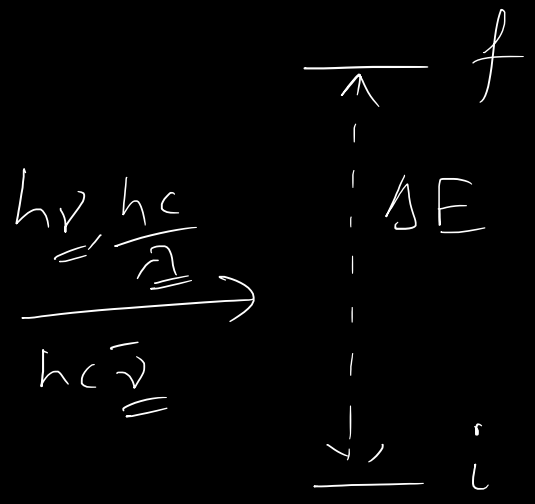


$\leftarrow E$

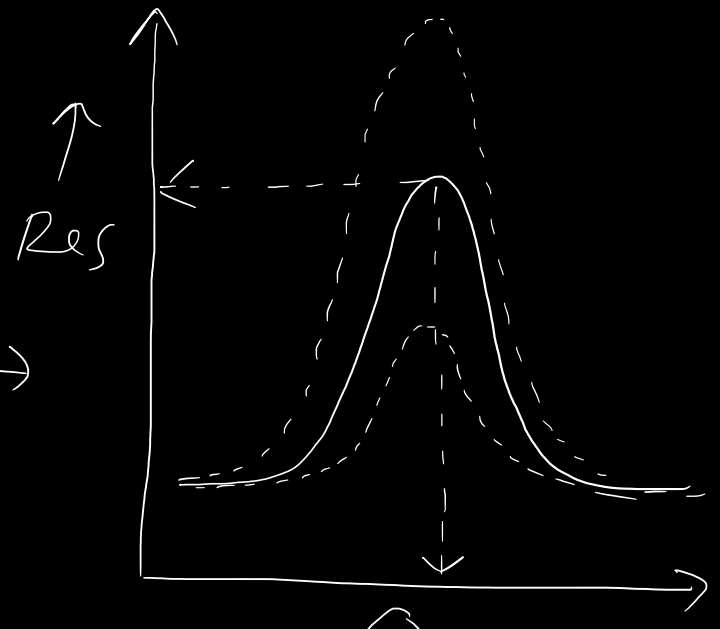
$\leftarrow \nu$

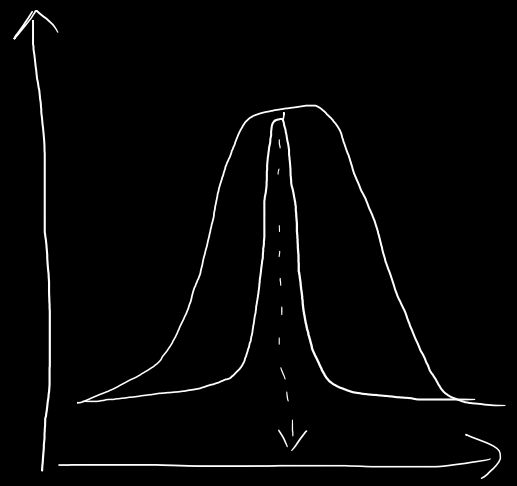
$\leftarrow \bar{\nu}$

$\leftarrow \lambda \rightarrow$



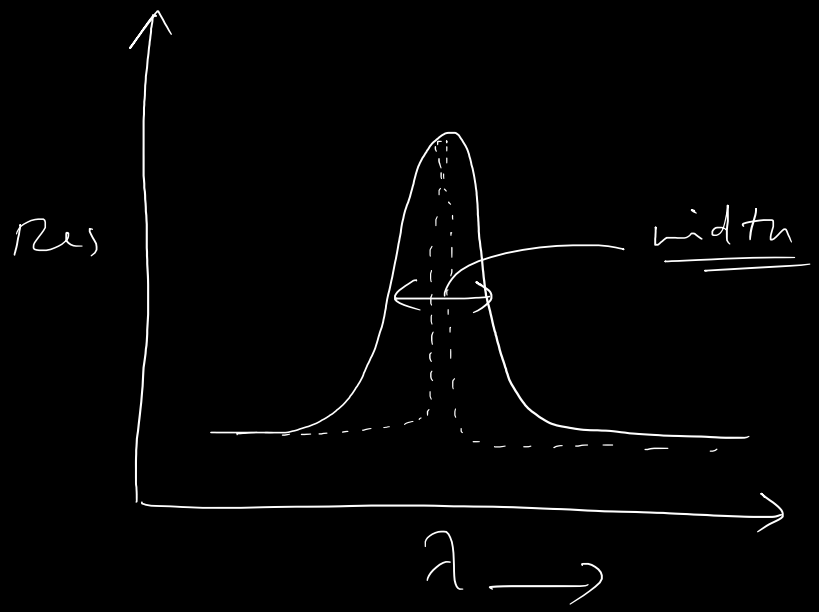
$$\Delta E = h\nu = \frac{hc}{\lambda} = hc\bar{\nu}$$

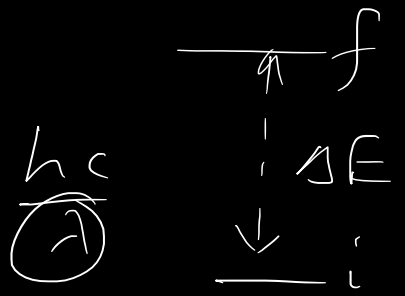




1. Peak Position
2. Peak Intensity : ??
3. Peak width : ⟶ ?

Peak width:





1. Doppler broadening :
2. Lifetime broadening :

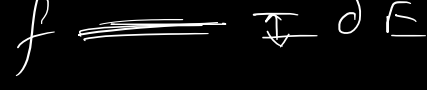
$$\delta t \, \delta E \simeq \hbar$$

δt : Lifetime of the excited state

δE : Uncertainty in the energy of the excited state.

$\delta t \, \delta E \gtrsim \hbar$





Lifetime broadening.

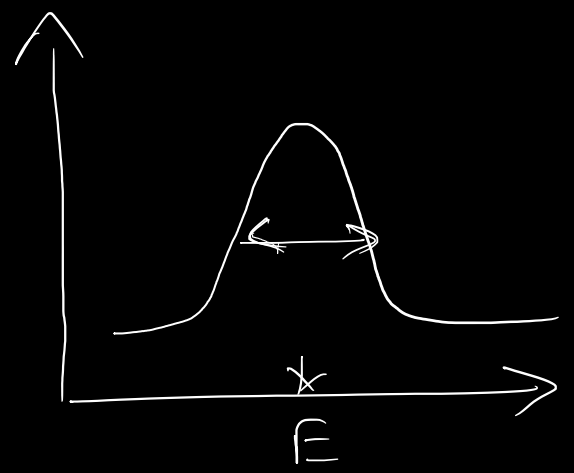
↓

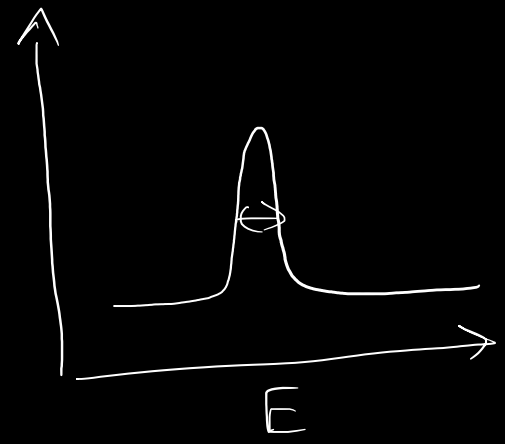
natural line width



$\delta t \rightarrow$ small

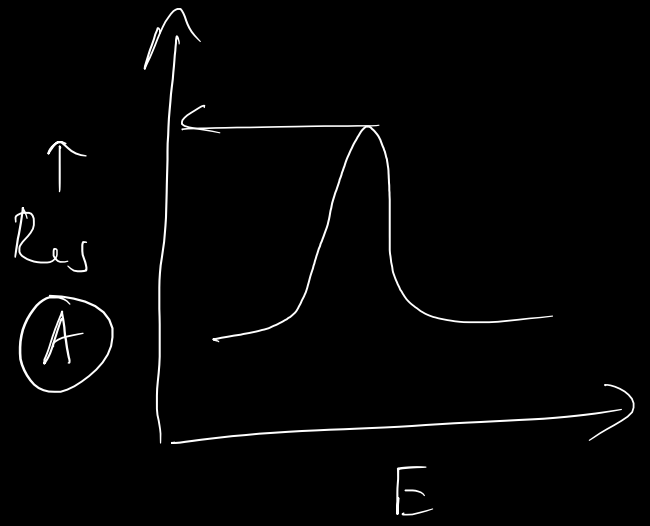
$\delta t \rightarrow$ large





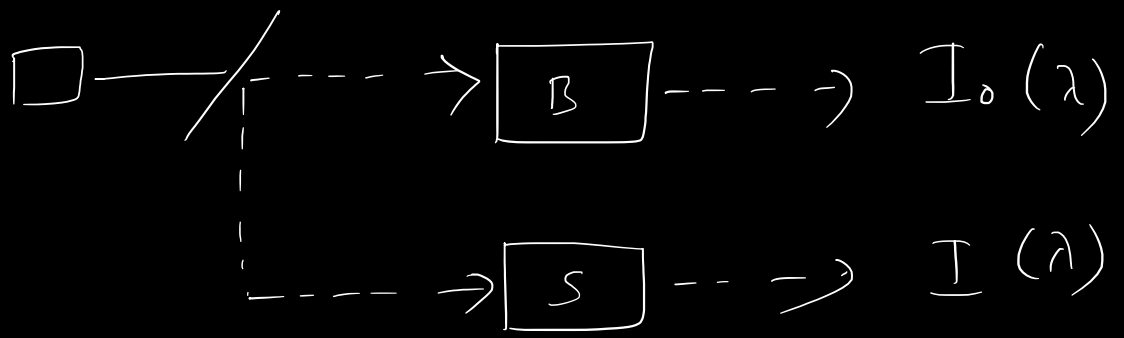
③ Peak Intensity

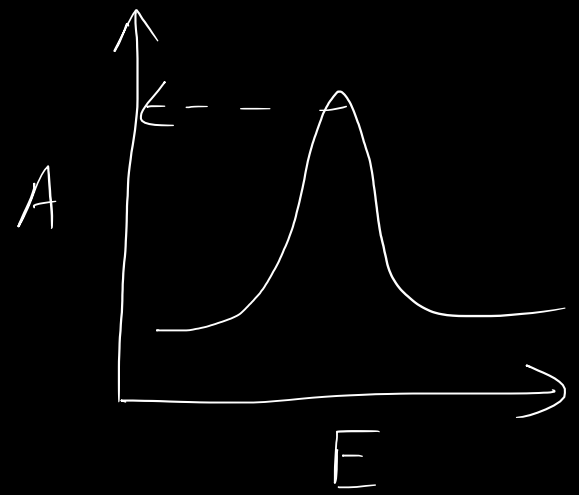




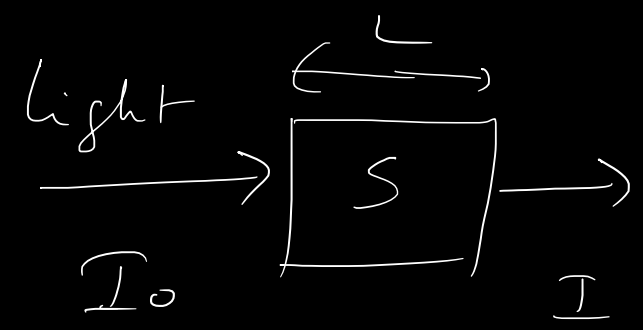
$$A = -\log_{10} \frac{I}{I_0} \qquad T = \frac{I}{I_0} \qquad \%T = \frac{I}{I_0} \times 100$$

$$A = -\log_{10} T$$





(Absorbance) $A = -\log_{10} \frac{I}{I_0}$



$A = \varepsilon \times C \times L$

Beer Lambert's law

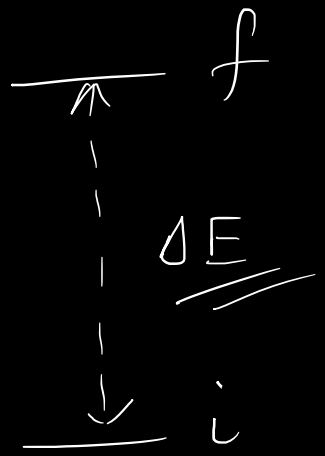
Molar absorption coefficient

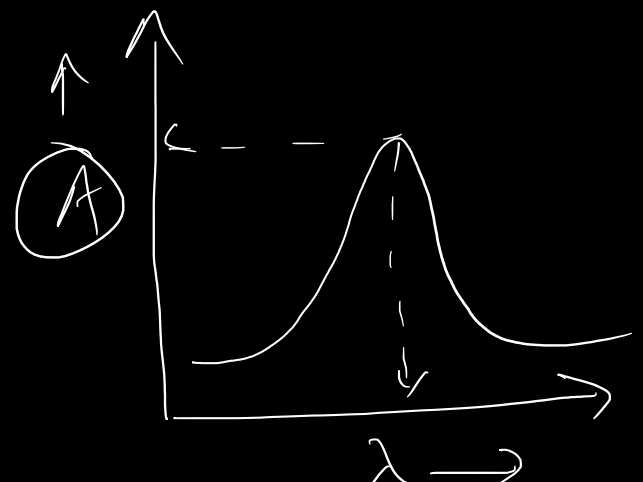
C → Concentration

L → length

$\varepsilon \rightarrow \int \psi_f^* \hat{\mu} \psi_i \, d\tau$ → $(dx\,dy\,dz)$

Transition dipole moment





Boltzmann Equation:

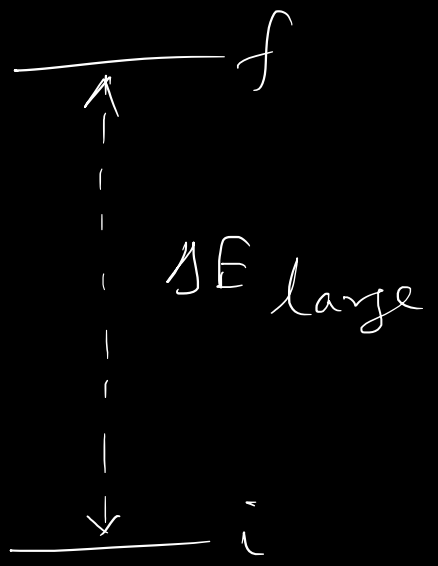


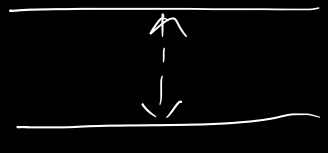
$$\frac{N_f}{N_i} = \square\, e^{-\Delta E/k_B T}$$

$$\frac{N_f}{N_i} = e^{-\Delta E/k_B T}$$

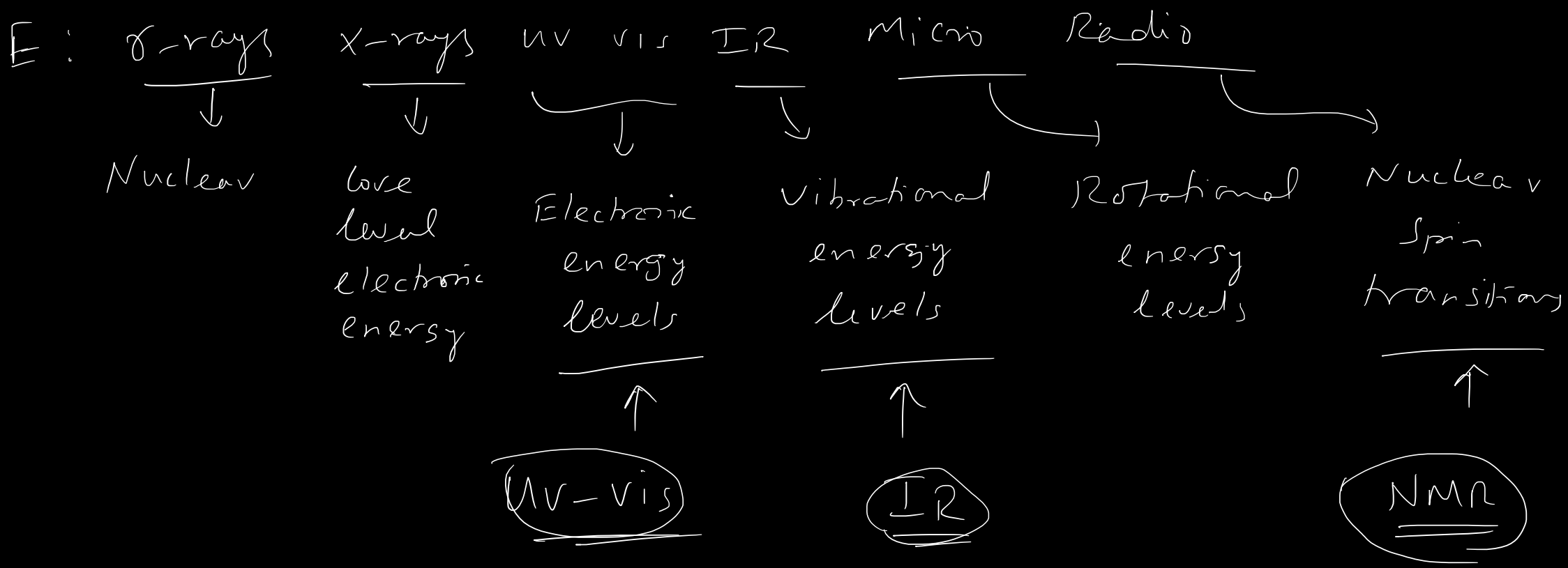
$k_B \rightarrow$ Boltzmann Constant

$1.381 \times 10^{-23}\ J\,K^{-1}$





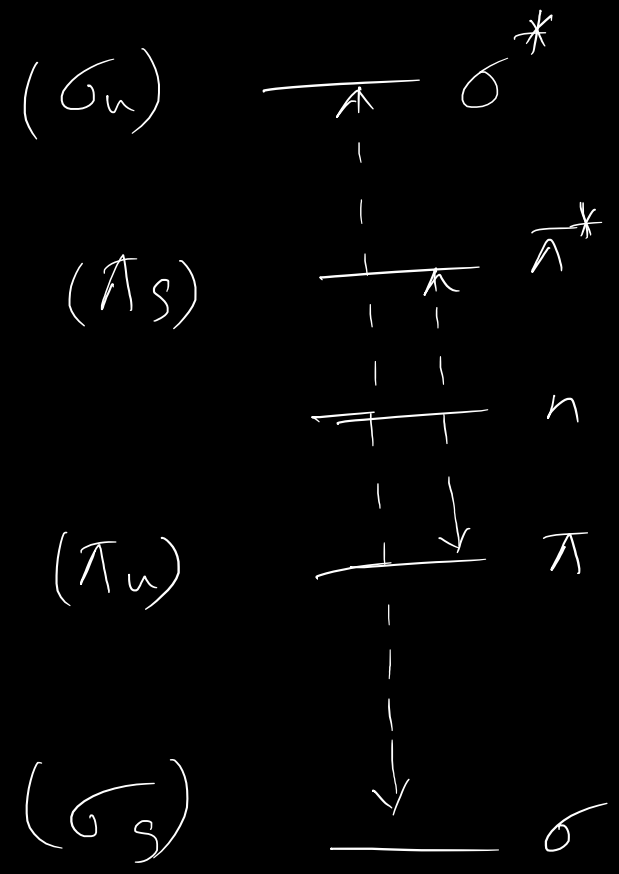
↳ $\frac{R}{N_A}$

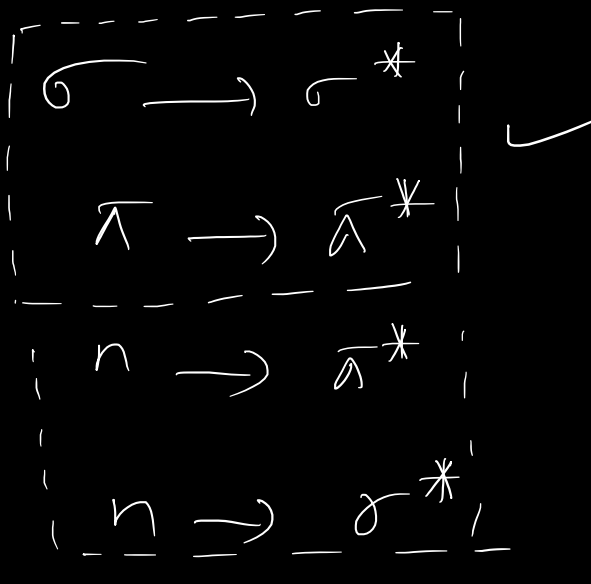

E : γ-rays X-rays UV VIS IR Micro Radio
Nuclear
Core level electronic energy
Electronic energy levels
Vibrational energy levels
Rotational energy levels
Nuclear Spin transitions
UV-VIS
IR
NMR

Electronic Spectroscopy (UV-Vis Spectroscopy)

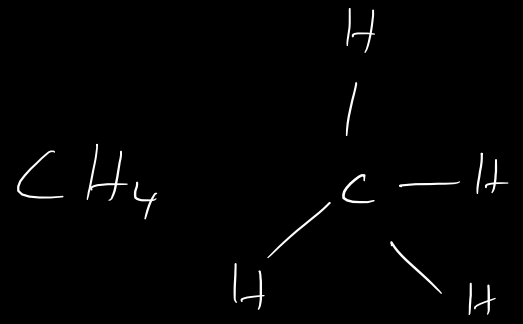
Organic

σ, π, n

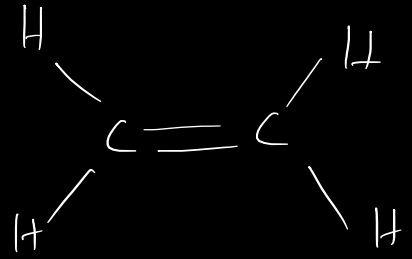




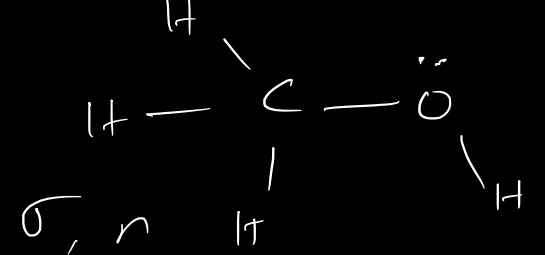
"Selection rules"



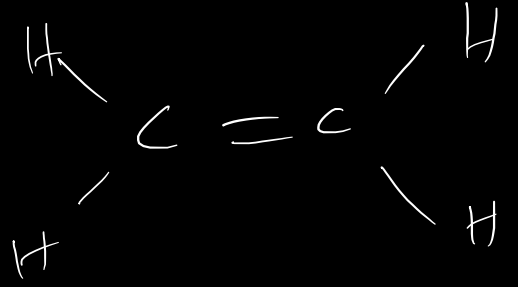
σ-bonds



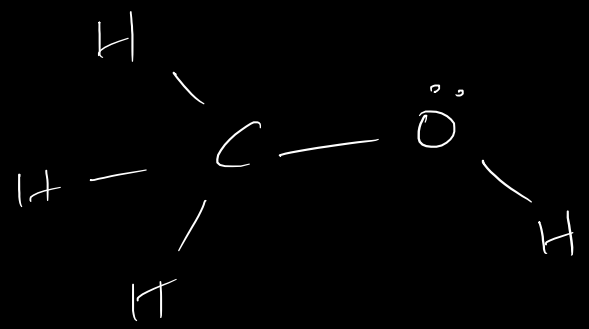
σ, π bond



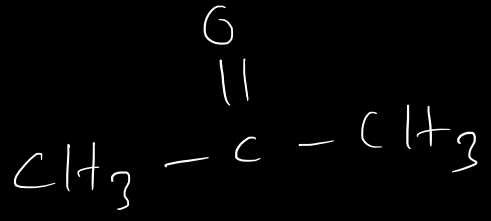
σ, n



$\sigma \rightarrow \sigma^*$
$\pi \rightarrow \pi^*$ $\Big] \rightarrow$



$\sigma \rightarrow \sigma^*$
$n \rightarrow \sigma^*$ $\Big] \rightarrow$



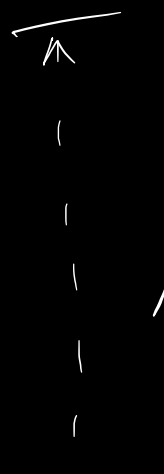
$\sigma \rightarrow \sigma^*$
$\pi \rightarrow \pi^*$
$n \rightarrow \pi^*$
$n \rightarrow \sigma^*$ $\Big] \rightarrow$

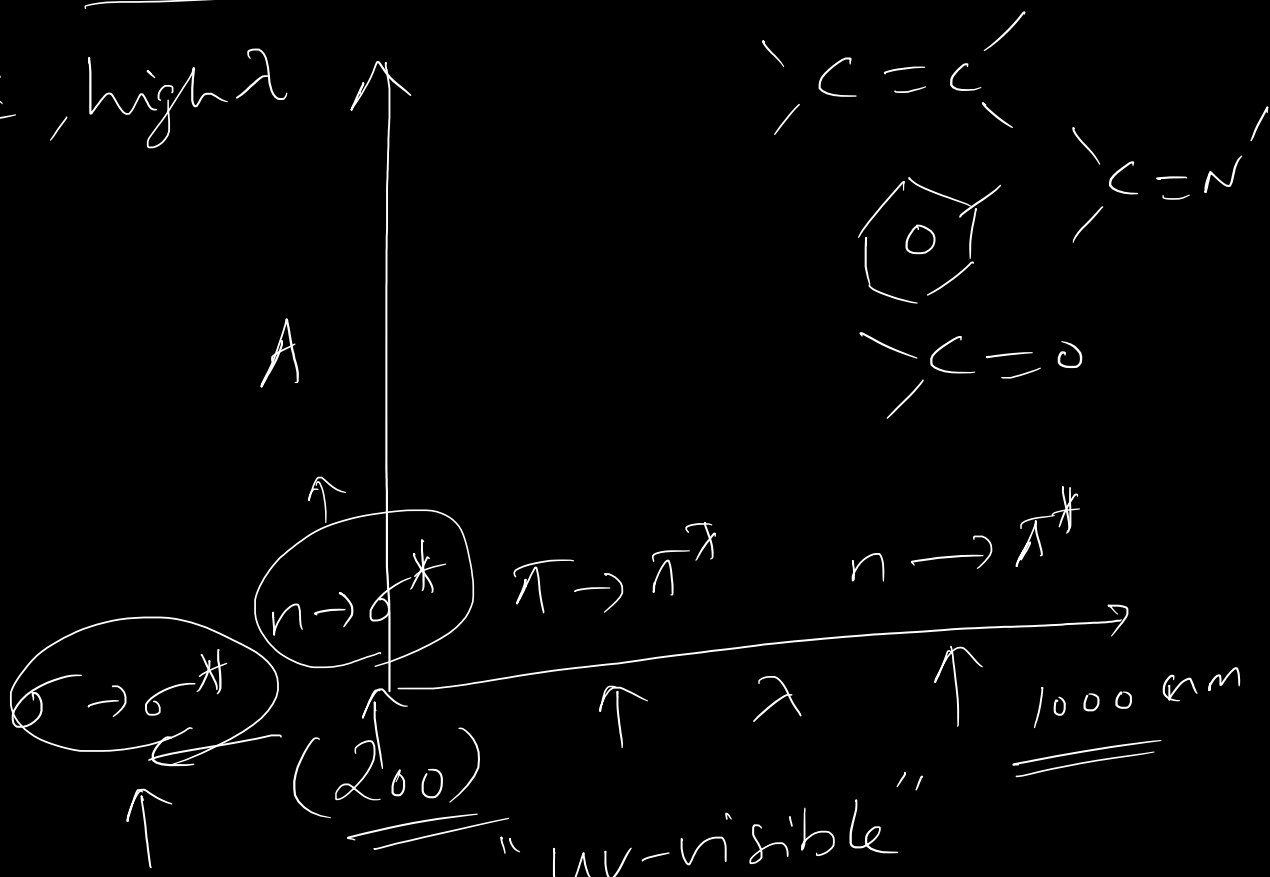
$\sigma \rightarrow \sigma^*$ — high ΔE, low λ

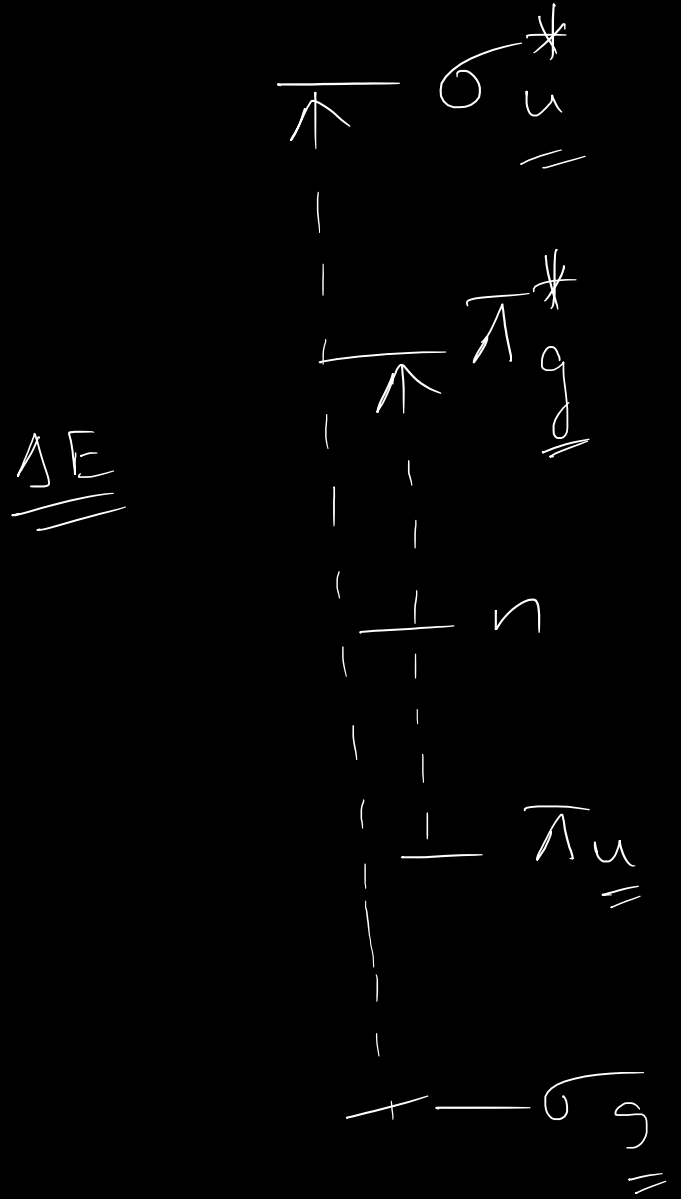
$\pi \rightarrow \pi^*$

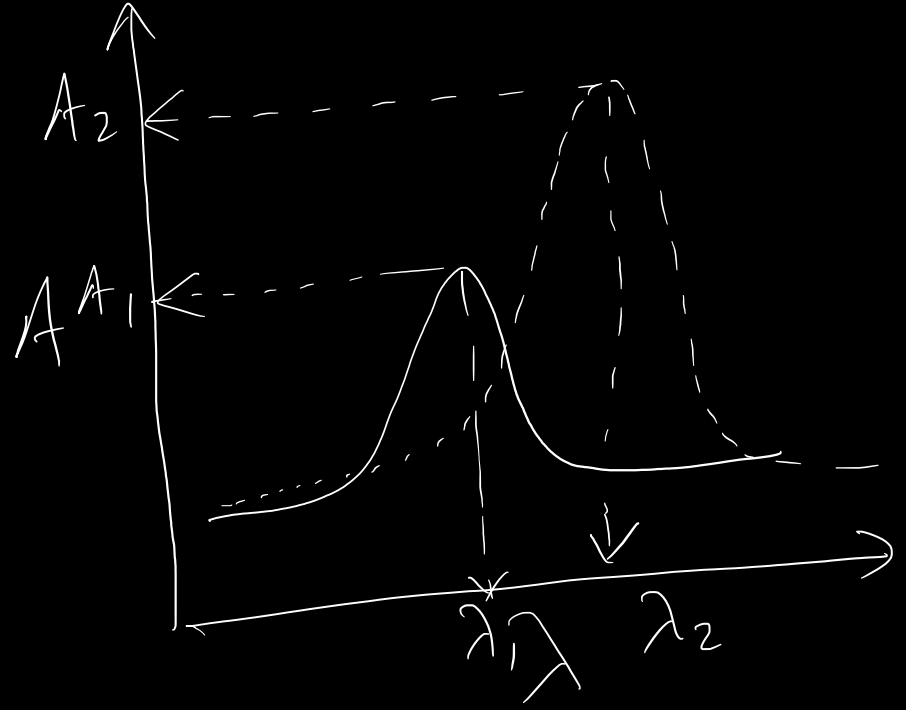
$n \rightarrow \pi^*$ — low ΔE, high λ → "chromophoric Groups"

$n \rightarrow \sigma^*$









$$A = \epsilon c l$$

$$\left.\begin{matrix} \sigma \rightarrow \sigma^* \\ \pi \rightarrow \pi^* \end{matrix}\right\} \text{allowed} \quad \epsilon \uparrow \text{ (strong)}$$

$$\left.\begin{matrix} n \rightarrow \pi^* \\ n \rightarrow \sigma^* \end{matrix}\right\} \text{not-allowed.} \quad \epsilon \downarrow \text{ (weak)}$$

intense

Acetone : (n-hexane)

n-hexane

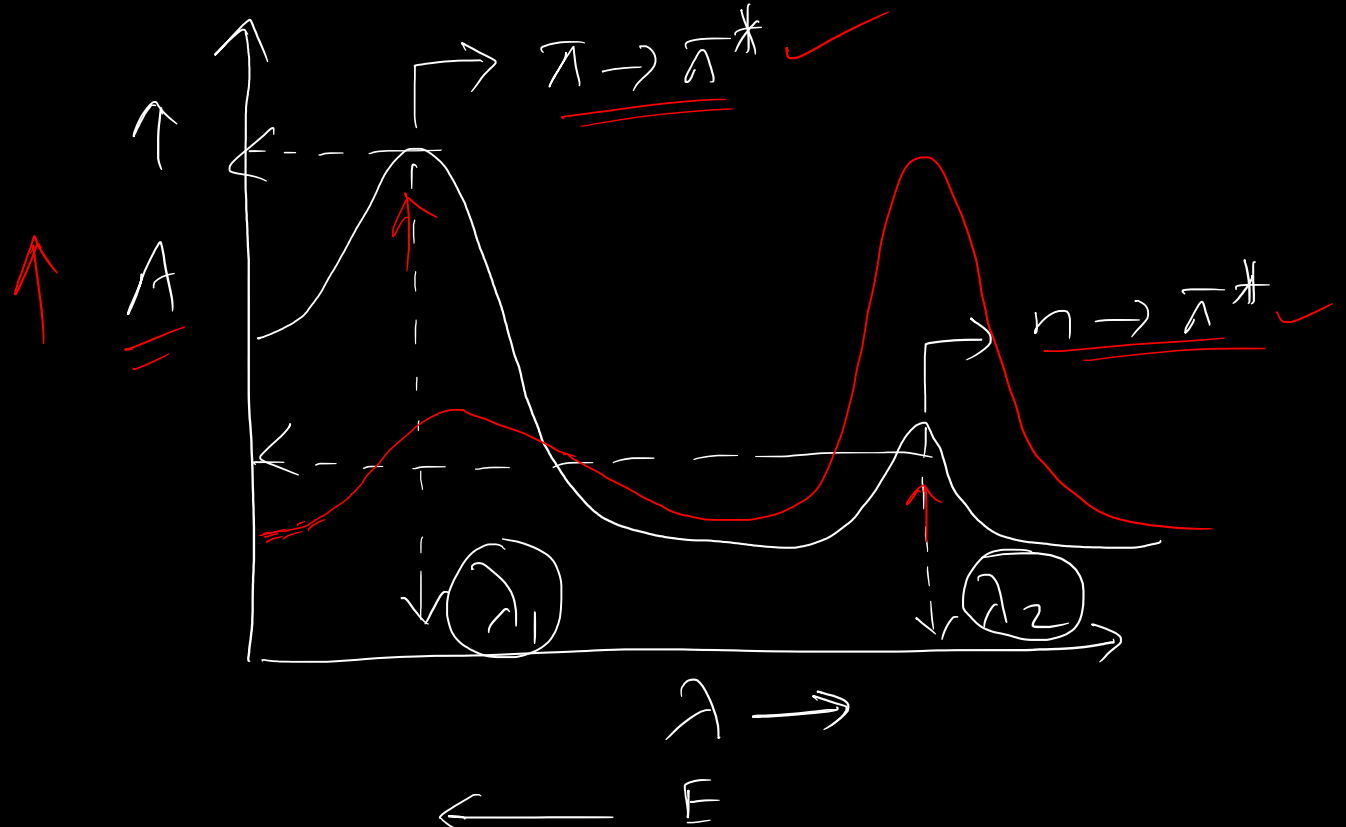
B ------> $I_0(\lambda)$

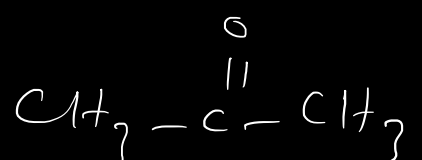
$\frac{I}{I_0} = T$

$A = -\log_{10} T$

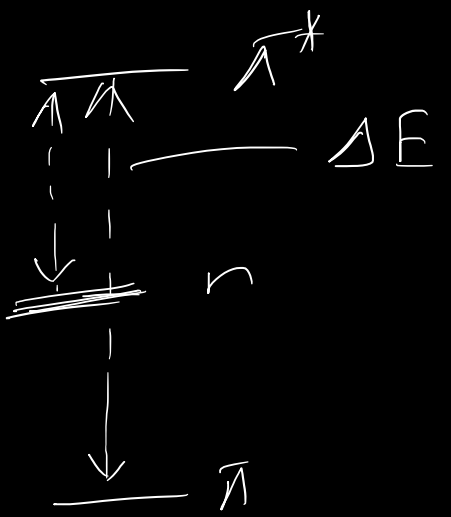
n-hexane + Acetone

S ------> $I(\lambda)$

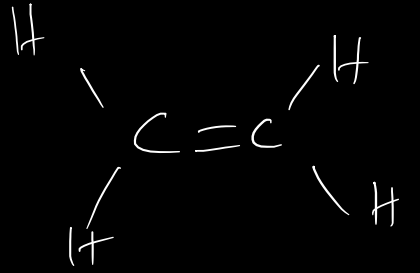




σ → σ*
π → π*
n → π*
n → σ*

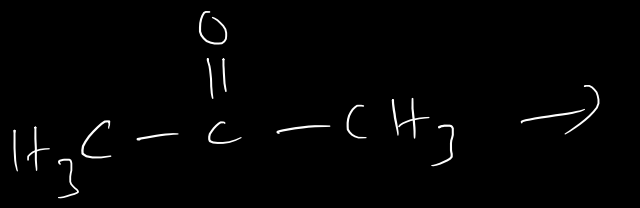


$CH_4 \longrightarrow \sigma \rightarrow \sigma^*$

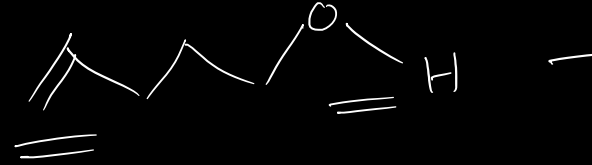


$\longrightarrow \sigma \rightarrow \sigma^*$, $\pi \rightarrow \pi^*$

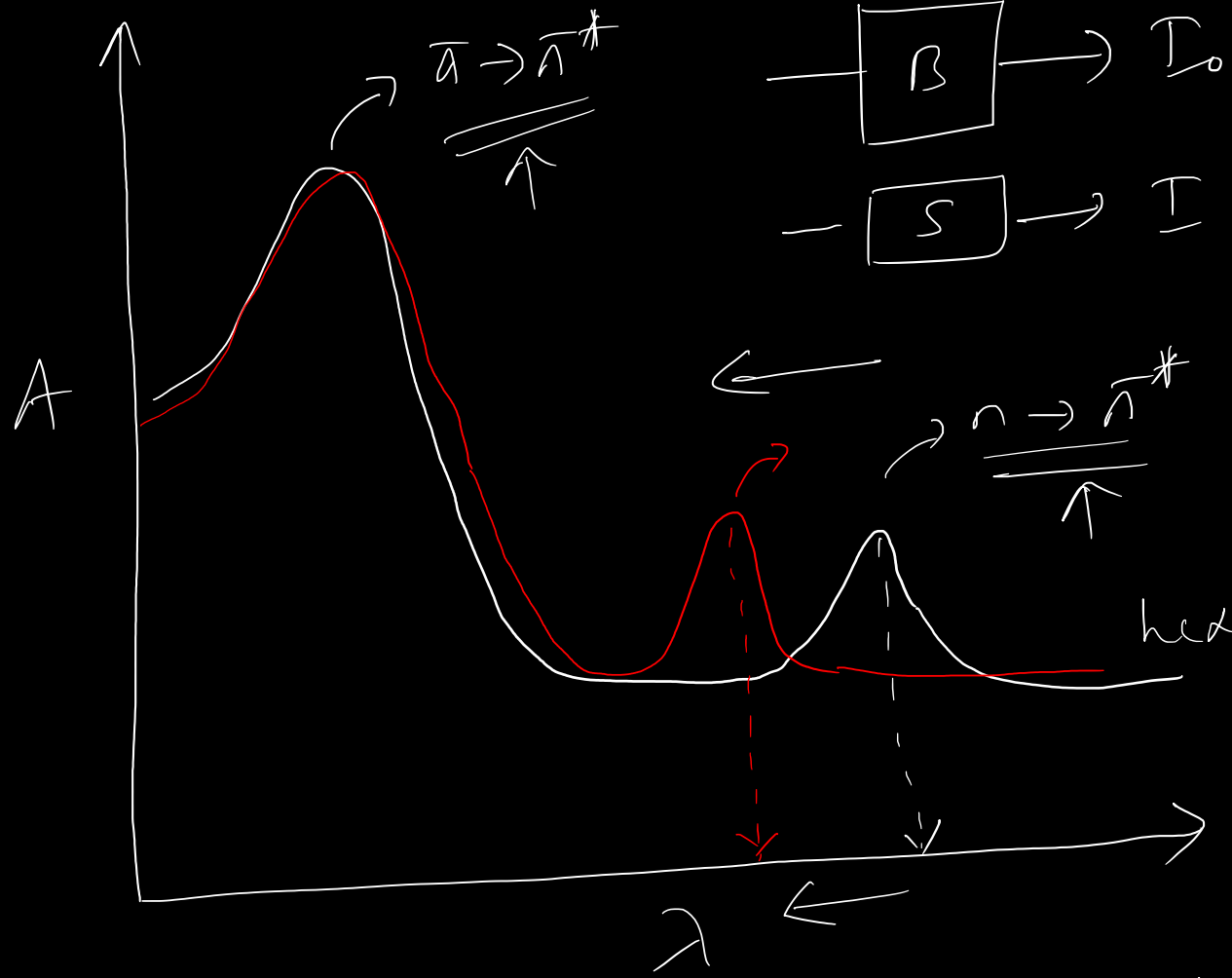
$CH_3-OH \longrightarrow \sigma \rightarrow \sigma^*$, $n \rightarrow \sigma^*$

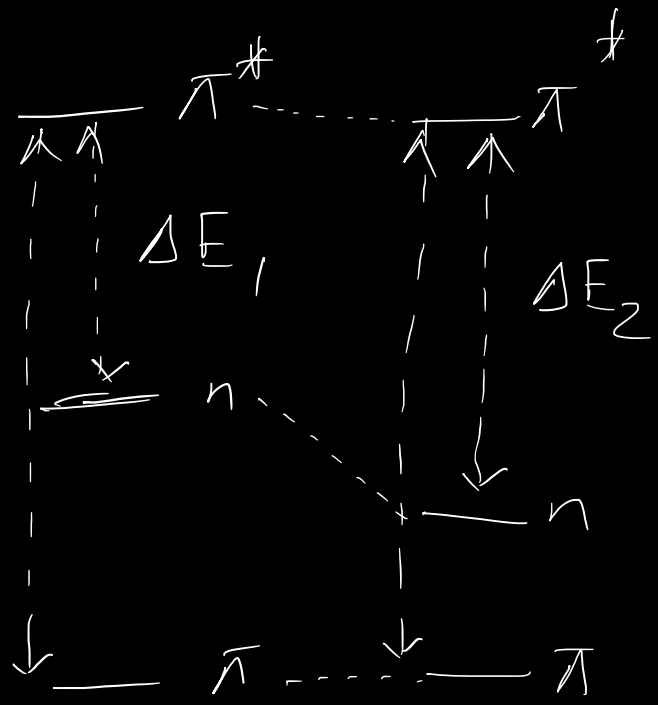


$\longrightarrow \sigma \rightarrow \sigma^*$, $\pi \rightarrow \pi^*$, $n \rightarrow \pi^*$, $n \rightarrow \sigma^*$



$\longrightarrow \sigma \rightarrow \sigma^*$, $\pi \rightarrow \pi^*$, $n \rightarrow \sigma^*$, ($n \not\rightarrow \pi^*$)

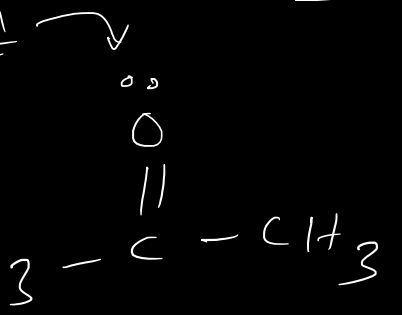




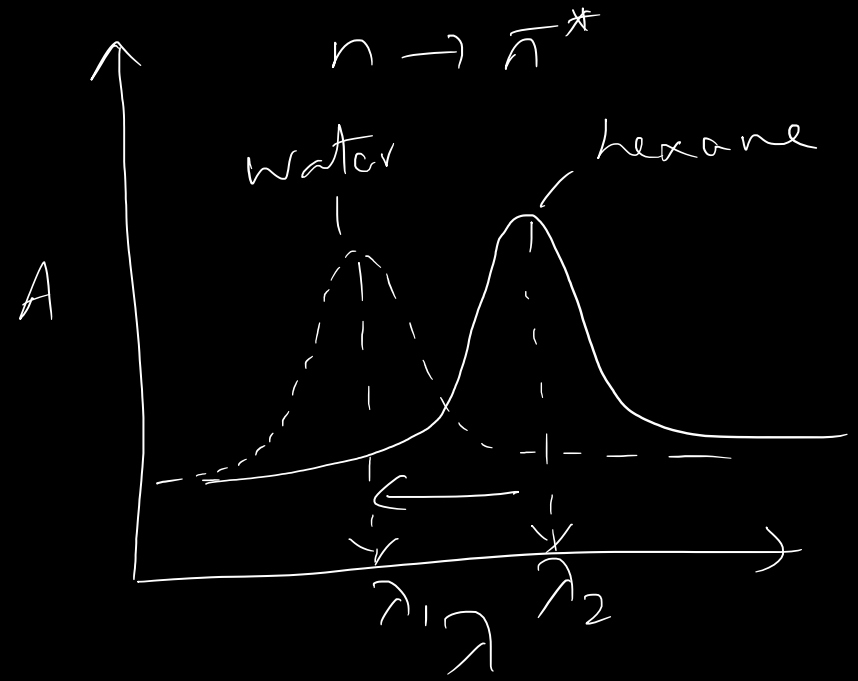
Acetone/ n-hexane

Acetone/ water

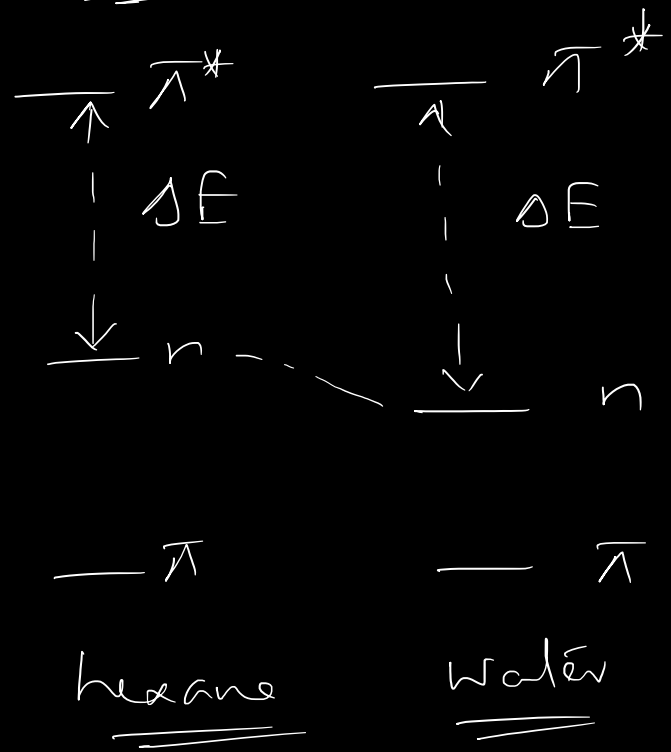
$n \rightarrow \pi^*$ → "blue shift"



Solvent effect $n \rightarrow \pi^*$ acetone



"Blue shift"



Effect of Conjugation

ethylene

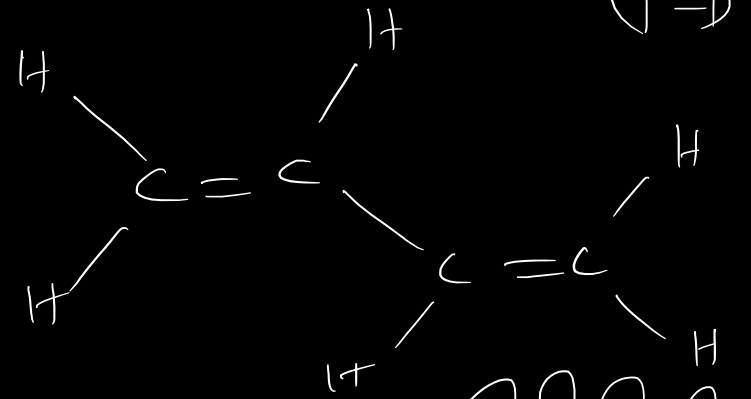


$\sigma \rightarrow \sigma^*$

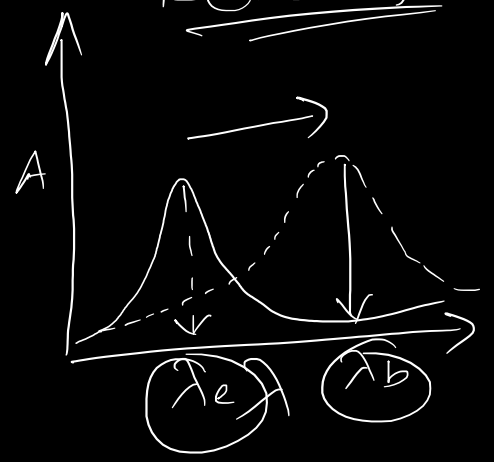
$\pi \rightarrow \pi^*$

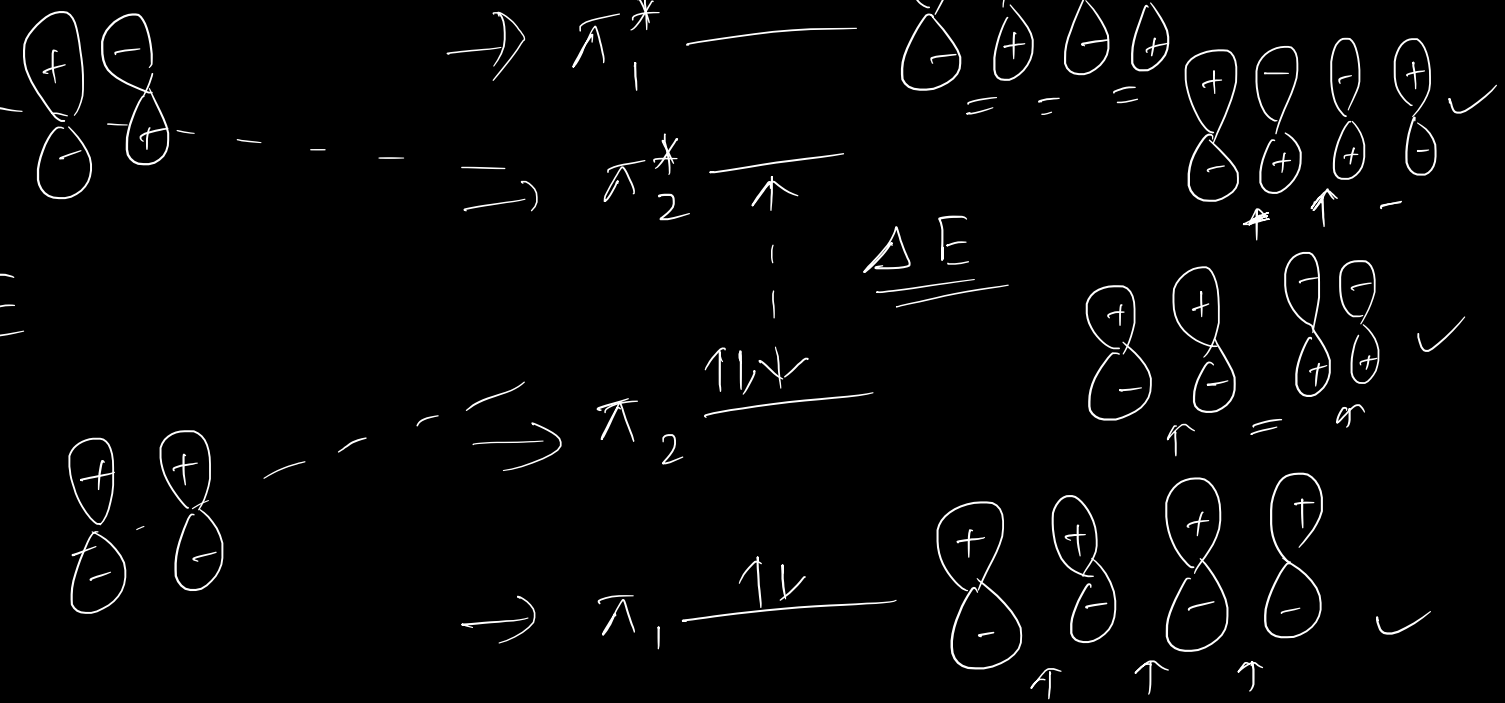
1,3-butadiene

(1 → box)



"Red Shift"





$$E = \frac{n^2 h^2}{8ma^2}$$

$n = 1$

$$\Delta E = \frac{(2n+1)h^2}{8ma^2} = \frac{3h^2}{8ma^2}$$

(éthylène)

$L \to a$

$L \to 2a$

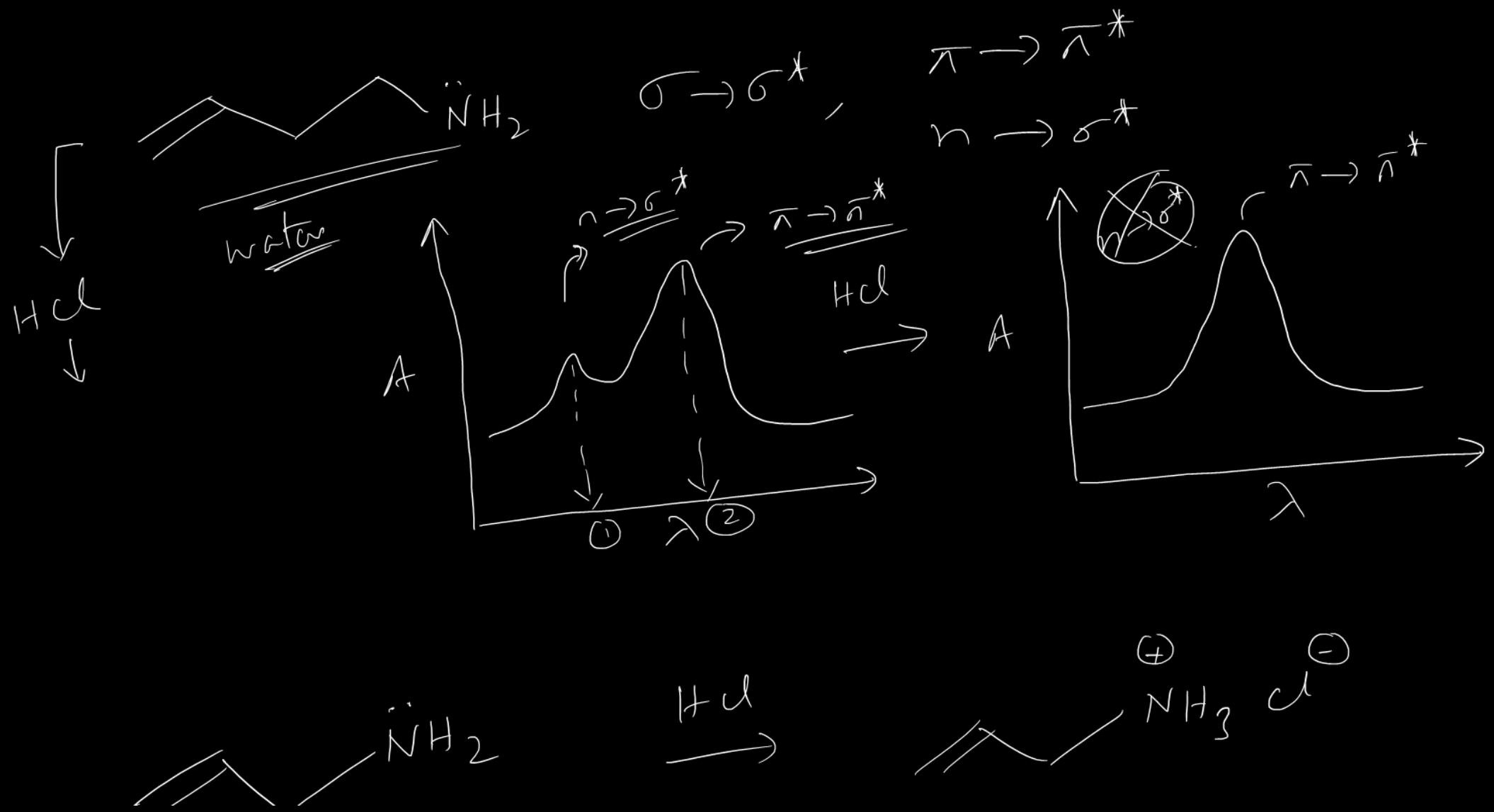
$(n = 2)$

$$\Delta E = \frac{(2n+1)h^2}{8m(2a)^2} = \frac{5h^2}{32ma^2}$$

(butadiène)

$$\Delta E_{\text{éthylène}} > \Delta E_{\text{butadiène}}$$

1, 3, 5 hexatriène


$\sigma \rightarrow \sigma^*$
$\pi \rightarrow \pi^*$
$n \rightarrow \sigma^*$
$\ddot{N}H_2$
water
HCl
$n \rightarrow \sigma^*$
$\pi \rightarrow \pi^*$
A
HCl
A
$\pi \rightarrow \pi^*$
σ^*
① λ ②
λ

$\ddot{N}H_2$
HCl
$\overset{\oplus}{N}H_3$ $Cl^{\ominus}$

Quantitative Analysis:

$$A = \underset{\uparrow}{\epsilon}\ \underset{\uparrow}{c}\ l$$

$C = 0$, $C = ?$

